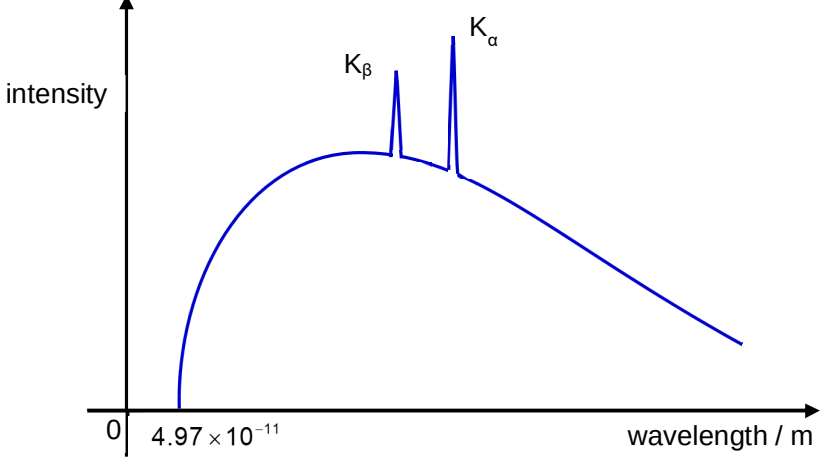


Qns	Answer	Marks
4(a)(i)	<u>electromagnetic radiation produced</u> whenever <u>charged particle</u> is <u>suddenly accelerated/decelerated</u> at the metal target (and wavelength depends on magnitude of acceleration) electrons hitting metal target have a <u>range/distribution of accelerations</u>	B1 B1
4(a)(ii)	<u>all</u> kinetic energy of <u>one electron</u> given up in <u>a single collision</u> to produce a single X-ray photon. $\lambda_{min} = hc/E$ so minimum wavelength depends only on energy of electrons Accelerating p.d are the same	B1 B1 B1
4(b)	at low voltages, the energy of electrons (25 keV) is <u>not sufficient to knock electrons out of the inner shells</u> of the tungsten atom	B1
4(c)	$E = \frac{p^2}{2m} = \frac{h^2}{2m\lambda^2}$ $(p = 8.54 \times 10^{-23} \text{ N s})$ $q_e V = \frac{h^2}{2m\lambda^2}$ $\lambda = \sqrt{\frac{h^2}{2mq_e V}}$ $= \sqrt{\frac{(6.63 \times 10^{-34})^2}{2(9.11 \times 10^{-31})(1.6 \times 10^{-19})(25000)}}$ $= 7.76 \times 10^{-12} \text{ m}$	C1 C1 manipulation / substitution A1

Qns	Answer	Marks
4(d)	 Working for λ_{min} $q_e V = \frac{hc}{\lambda_{min}}$ $\lambda_{min} = \frac{hc}{q_e V}$ $= \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{(1.6 \times 10^{-19})(25000)}$ $= 4.97 \times 10^{-11} \text{ m}$	B1 (shape) B1 (K_α to right of K_β, labelled) B1 (value of λ_{min})

5(a)(i)	work done per unit charge where electrical energy is converted into	B1
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